SQL SATURDAY · BOSTON · OCTOBER 3, 2026

Intro to PowerShell Automation for the SQL Server DBA

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THE PROBLEM

Our data estates are growing. Our teams are not.

- More instances, more versions, more places they live
- The same finite number of DBAs looking after them
- Clicking through SSMS four times does not scale to forty

> If you only do a thing once, do it by hand. If you will do it again, script it.

THE IDEA

PowerShell does not make one task faster. It multiplies.

```
total time  =  minutes per server  ×  servers  ×  times per year  +  cost of errors
```

- **Once** — get the minutes down, on one server
- **Everywhere** — take the *servers* term out
- **Forever** — take the *times per year* term out
- **Proven** — go after the *cost of errors*

AGENDA

Four acts

- **Once** — do the task once, by hand
  - A command returns objects, not text
- **Everywhere** — the same work, whole estate
  - One variable is the only thing that changes
- **Forever** — stop being the trigger
  - Collect it, keep it, schedule it
- **Proven** — prove it actually worked
  - The part nobody was doing at all

FOLLOW ALONG

Four servers on your laptop

CONTAINER	VERSION	CONNECT WITH
`sqlsat2017`	SQL Server 2017	`localhost,14317`
`sqlsat2019`	SQL Server 2019	`localhost,14319`
`sqlsat2022`	SQL Server 2022	`localhost,14322`
`sqlsat2025`	SQL Server 2025	`localhost,14325`

Docker Desktop · PowerShell 7 · `Install-Module dbatools` · ~12 GB disk

Pull the images at home — conference wifi will not thank you.

ACT 1

Once

- Your first command — `Verb-Noun`, and why it returns objects
- Arguments — `-Database Sales` is `@Database = 'Sales'`
- Variables — the seam where one server becomes forty
- Piping — `Where`, `Sort`, `Select`, in that order
- Modules — where `Get-DbaDatabase` actually came from

ONCE

A command returns objects, not text

```powershell
# "Which databases are in FULL recovery but have never had a log backup?"
Get-DbaDatabase -SqlInstance $Target -ExcludeSystem @Lab |
    Where-Object { $_.RecoveryModel -eq 'Full' -and
                   $_.LastLogBackup -lt [datetime]'1900-01-01' } |
    Select-Object Name, RecoveryModel, LastFullBackup, LastLogBackup
```

No parsing. No string handling. `Where-Object` *is* `WHERE`*,* `Sort-Object` *is* `ORDER BY`*,* `Select-Object` *is* `SELECT` *— you already know this query.*

ACT 2

Everywhere

- One line, four servers — the whole session in one script
- Fleet inventory — what do we actually have?
- Fleet backup gap sweep — is everything backed up?
- Fleet maintenance rollout — the half-day-per-server one
- Fleet job audit — six months later, four different answers
- Fleet job fix — the same correction, everywhere at once

EVERYWHERE

One variable is the entire trick

```powershell
# Act 1 ended here
$Target  = 'localhost,14317'

# Change one line
$Targets = 'localhost,14317','localhost,14319','localhost,14322','localhost,14325'

Get-DbaDatabase -SqlInstance $Targets -ExcludeSystem @Lab |
    Select-Object SqlInstance, Name, RecoveryModel, SizeMB, LastFullBackup |
    Sort-Object   SqlInstance, Name
```

No loop. No `ForEach`. No rewrite — `-SqlInstance` takes an array.

EVERYWHERE

The fleet disagrees on purpose

- **2017** — no backups at all
- **2019** — `Sales` only
- **2022** — everything, healthy
- **2025** — everything, but `Sales` is `COPY_ONLY`

The last one is the interesting one. It looks like a real full backup until you check `LastFullBackupIsCopyOnly`.

ACT 3

Forever

- Load data to a database — keep what you collected
- Scheduled report — Task Scheduler, and the report nobody has to remember to run

FOREVER

A fact, or a decision

A CSV tells you what is true right now.
A table tells you what **changed**.

- "Sales is 40 GB" is a fact
- "Sales grew 12 GB this month" is a decision

You cannot get the second one without having kept the first.

ACT 4

Proven

- Test the backups — restore, `CHECKDB`, drop, repeat
- Prove maintenance ran — deploying is not the same as working
- The scoreboard — measured live, on the servers you just built

PROVEN

A backup you have never restored is a hope

- Most shops do not do restore testing **badly** — they do not do it **at all**
- By hand it is most of a day per server, so nobody has the days
- `Test-DbaLastBackup` restores it, runs `CHECKDB`, and drops it

> Rolling maintenance out is not the same as it working. Act 2 deployed it.
> This closes the loop.

PROVEN

The scoreboard

Run `4 - Proven\3 - The scoreboard.ps1` — every number measured in this room, on the servers you just watched get built.

TASK	SCRIPT	BY HAND
Inventory the estate	seconds	6 min × 4
Check every backup	seconds	10 min × 4
Verify maintenance ran	seconds	15 min × 4
Restore-test the backups	seconds	**never**

TAKE IT HOME

The toolbox — 15 scripts

- Discovery and inventory
- Backups and recovery
- Integrity and corruption
- Security and access
- Configuration and drift
- Agent jobs
- Capacity and growth
- Performance
- Migration and refresh
- Windows and server

THANK YOU

Questions?

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Slides, scripts, demos and resources:

https://bit.ly/4m05kBy

Everything here is free:

```
Install-Module dbatools -Scope CurrentUser
```

